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Author(s): Lawrence B. Pulley

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A GENERAL MEAN-VARIANCE APPROXIMATION TO
EXPECTED UTILITY FOR SHORT HOLDING PERIODS

*Lawrence B. Pulley**

The mean-variance model is precisely consistent with the expected utility hypothesis only in the special cases of normally distributed security returns or quadratic utility functions. There is little evidence, however, that security returns follow normal distributions (see [13] for references) and quadratic preferences can be shown to generate implausible results, exhibiting increasing absolute risk aversion in the Pratt [11]-Arrow [1, 2] sense and displaying negative marginal utility after some finite wealth level. In addition, Hakansson [4] has shown that single-period, mean-variance-efficient portfolios can have disastrous consequences over time--even when return distributions are stationary. Such criticisms of the mean-variance approach within the Von Neumann-Morgenstern framework have prompted several writers to suggest that investors maximize the expected value of utility functions with more "realistic" properties, while others have criticized the single-period focus of the model. One popular alternative utility function is the logarithmic function which exhibits decreasing absolute risk aversion and (conveniently) leads to myopic decision processes through time (i.e., investors treat each period as if it were the last, basing investment decisions on that period's wealth and return distributions only [8, 4]). (Other utility functions with constant relative risk aversion--such as the power function--also imply myopic decision rules within a multiperiod setting.)

As is well known, the two-parameter, mean-variance model can be expected to produce asymptotically valid approximations to other utility functions when risks are small (as implied by Samuelson's [15] "compact" distributions) or when the length of the trading interval is close to zero [9]. Furthermore, Levy and Markowitz [7] have empirically demonstrated the validity of the approximations using historical security-return distributions for realistic holding periods. Theirs is but one of many attempts [12, 17, 18] to reconcile the mean-variance approach

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with the Von Neumann-Morgenstern, expected-utility hypothesis. There are, of course, numerous possible two-parameter approximations, the functional forms of which traditionally depend in a complicated way on the utility function being approximated. Thus, any attempt to apply the mean-variance approximations in a positive context would, although reducing analytical complexity and trimming computing expense, still require knowledge of the actual utility functions of the individuals involved. The purpose of this paper is to derive a general mean-variance formulation which can be used to approximate a wide class of utility functions in the absence of complete knowledge of the specific form of those functions and which does not require restrictive assumptions concerning the distributions of security returns. The validity of the approximations will be empirically examined in a portfolio-selection context using historical security-return distributions for monthly and semiannual periods. The mean-variance formulation which will be developed is based on Taylor-series approximations and can be represented as:

$$(1) \quad G[E(U)] = \mu - \alpha\sigma^2$$

where

μ is the mean rate of return,

σ^2 is the variance of the rate of return r , and

α is a constant equal to $1/2$ times the Pratt-Arrow relative-risk-aversion coefficient ($RRA = -YU''(Y)/U'(Y)$) evaluated at $r=0$.

In Section I of the paper expression (1) above is derived and discussed. In the second section actual portfolios maximizing logarithmic, power, and exponential utility functions are calculated and compared with portfolios maximizing expression (1). Summary and concluding remarks are contained in Section III.

I. Rationale

Consider the general Von Neumann-Morgenstern utility function defined on final wealth $U(\tilde{Y})$ where $\tilde{Y} = w(1+r)$ with w being initial wealth and r being the rate of return. (Let μ and σ^2 represent the mean and variance of r .) We are interested in the limiting case, i.e., the case in which the length of the holding period, T , goes to zero. Assume that $\mu = o(T)$, $\sigma^2 = o(T)$, and that the third absolute moment is $o(T)$.¹ Then, with a few minor restrictions on distributions and utilities, Ohlson [9] has shown that the Taylor series expansion of

¹For any sequence $a(T)$, define $a(T) = o(T^n)$ iff $\lim_{T \rightarrow 0} |T^{-n}a(T)| = c > 0$, and $a(T) = o(T^n)$ iff $\lim_{T \rightarrow 0} |T^{-n}a(T)| = 0$.

$\tilde{U}(Y)$ about mean wealth $v (=w(1+\mu))$ yields:

$$(2) \quad E[U(Y)] = U(v) + U''(v)w^2\sigma^2 + o(T).$$

However, based on our assumptions and Ohlson's results, we get by expanding about $\mu = 0$:

$$(2a) \quad U(v) = U(w) + U'(w)w\mu + o(T)$$

$$(2b) \quad U''(v) = U''(w) + o(T^0).$$

Substituting (2a) and (2b) into (2) and noting that, since $\sigma^2 = o(T)$, it follows that $o(T^0) \times \sigma^2 = o(T)$, one finds

$$(3) \quad E[\tilde{U}(Y)] = U(w) + U'(w)w\mu + U''(w)w^2\sigma^2 + o(T)$$

which is equivalent as an objective function to

$$(4) \quad \begin{aligned} G[E(\tilde{U}(Y))] &= \mu + 1/2 U''(w)/U'(w)w\sigma^2 \\ &= \mu - 1/2 \text{RRA} \Big|_{r=0} \sigma^2 \\ &= \mu - \alpha\sigma^2. \end{aligned}$$

Thus, the investment problem of selecting the portfolio which maximizes expected utility reduces, for small r , to the problem of finding the portfolio maximizing expression (4) above. Since the coefficient of relative-risk aversion (RRA) is evaluated at $r=0$, α is a constant with respect to the actual computations. Incidentally, expression (4) is very similar to the model used in Donald Farrar's important contribution, *The Investment Decision under Uncertainty* (Prentice-Hall (1962). See page 21).

Before calculating and comparing the portfolios which maximize general logarithmic, exponential, and power utility functions with those maximizing expression (4) above, it is, perhaps, beneficial to discuss some of the implications of the formulation. Initially it should be noted that expression (4) is an intuitively pleasing and plausible result--fusing the simplest possible mean-variance formulation with the Pratt-Arrow concept of risk aversion. It is the relative-risk-aversion coefficient which, in fact, determines the weight assigned to an additional unit of variance. Furthermore, if the mean-variance expression

in (4) above is valid, utility functions with identical risk-aversion coefficients should yield identical optimal portfolios.

The mean-variance approximation as given in expression (4) could have significant impact on normative portfolio models. If the approximation is a good one, mutual funds and other institutional investors could, for example, closely adhere to the normative rule of maximizing the geometric-mean return (or, equivalently, maximizing expected logarithmic utility) very inexpensively since the optimal portfolios from (4) are easily computed using any one of a number of quadratic programming algorithms. More importantly, however, by establishing investment funds based on expression (4) for a range of relative-risk-aversion-coefficient values, a mutual fund could assure itself of selecting portfolios which maximize utility for a wide class of individual investors having different utility functions and wealth levels, regardless of the actual form of their utility functions.

If expression (4) provides a valid approximation, the continuing debate in positive analysis concerning which utility function best describes the preferences of rational investors (see, for example, [14]) is essentially moot. If empirical studies can be devised to estimate risk-aversion coefficients, the above mean-variance formulations could be employed to describe or predict behavior for all expected utility maximizers in that risk class, regardless of which "utility function" they possess.

Of course, all of these results are valid only to the extent that expression (4) provides a good approximation to the expected value of more general utility functions. In the next section we will examine the goodness of the approximation for various utility functions and actual security-return distributions.

II. Empirical Results

Implicit in the theoretical discussion of the previous section and explicit in subsequent empirical investigations is the assumption of myopic maximization of expected utility with monthly (or semiannual) holding periods. For investors in a multiperiod setting, this generally requires that the investor's objectives for wealth generate a myopic decision function defined on monthly (semiannual) holding periods, requiring serially-independent security returns. As Mossin [8] has shown, however, investors with multiperiod preferences described by logarithmic or power utility functions would optimize myopically. Also, Breiman [3] and Thorp [16] have proved that investors maximizing the geometric-mean return would myopically maximize the expected logarithm of wealth each period--even if return distributions display serial dependence.

Recall Expression (4)

$$(4) \quad G[E[U(\tilde{Y})]] = \mu - \alpha \sigma^2 = \mu - 1/2 \text{ RRA} \Big|_{r=0} \sigma^2.$$

In Table 1 the risk-aversion coefficients are presented for general logarithmic, power, and exponential utility functions.²

TABLE 1

Formulation	Absolute Risk Aversion (r=0)	Relative Risk Aversion (r=0)	Relevant Parameter Range
Logarithmic $U(\tilde{Y}) = \ln[w(1+w+b_1)]$	$1/[w(1+b_1)]$	$1/(1+b_1)$	$b_1 > -1$
Power $U(\tilde{Y}) = [w(1+r)]^{b_2}$	$(1-b_2)/w$	$1-b_2$	$0 > b_2 > 1$
Exponential $U(\tilde{Y}) = 1 - \exp^{-b_3 w(1+r)}$	b_3	$b_3 w$	$b_3 > 0$

As implied by expression (4) and the values in Table 1, the optimal portfolios of the logarithmic and power utility functions are independent of initial wealth. This is not true for the exponential function; however, since the relative-risk-aversion coefficient is equal to $b_3 w$, w can be set equal to one and all parameter changes can be represented as changes in b_3 alone. (In our portfolio simulations initial wealth, w , will be assigned the value one.)

Using monthly security returns (1960-1976)³ for three arbitrarily selected samples having ten securities each, the portfolios maximizing expected utility for the general logarithmic, power, and exponential utility functions were computed for various values of the relevant parameters. Using actual returns, the expected utility-maximizing portfolio is calculated as that portfolio maximizing:

(5)
$$E[U(\tilde{Y})] = \sum_{t=1}^{204} U(w \sum_{i=1}^{10} a_i (1+r_{it}), b_k)$$

where

- t refers to the monthly observation,
- a_i is the proportion of wealth invested in security i ,
- r_{it} is the rate of return on security i in period t , and
- b_k is the utility-function-specific parameter value.

The problem in expression (5)--implying the maximization of expected utility over

²These utility functions are used because of their position in the literature [4, 5, 6, 8, 10, 12, 13, 14].

³Source: Chicago Center for Research in Security Prices monthly return tape.

the exact empirical distribution--was solved using the modified Frank-Wolfe algorithm described in Ohlson and Ziemba [10]. With the only constraint being that the portfolio weights must sum to one, the algorithm is very easily solved. The a_i 's in all simulations are constrained to be greater than or equal to zero, i.e., short sales are ruled out. The optimal a_i 's obtained by maximizing expected utility as given in expression (5) will be referred to as "utility-optimal" weights. The portfolio weights maximizing expression (4) with appropriate values for alpha were also computed (referred to as "mean-variance-optimal" weights) using a standard quadratic-programming algorithm. The ratios of expected utility for each utility function using the mean-variance-optimal weights to expected utility using the utility-optimal weights are reported in Table 2. The mean-variance-optimal weights yield values of expected utility virtually identical to those for the utility-optimal weights.

It is also the case that the optimal weights for the utility functions and for their mean-variance counterparts are extremely similar in all comparisons. Also, as expected, portfolio weights across utility functions are quite similar when parameter values are selected to yield the same values for the local coefficient of relative-risk aversion. Table 3 presents evidence on these last two points. In Table 3 the portfolio weights for the relevant utility functions and their mean-variance counterparts are reported for parameter values yielding similar local risk-aversion coefficients. (The relevant parameter values were calculated using column 2 of Table 1 and assuming $w=1$.) Looking across the rows in Table 3 we see that, when parameters are chosen so that the RRA's are equal, the optimal portfolios are almost the same. In addition, for each utility function, the mean-variance-optimal weights are very close to the utility-optimal weights. And, as the risk-aversion coefficient (and α) gets smaller, the individual, of course, becomes less risk-averse and the optimal portfolio becomes more heavily concentrated in securities with the highest mean returns. In fact, for relative-risk-aversion coefficients on the order of .1 or .2, all wealth would be invested in the security with the highest mean return. For large values of relative-risk-aversion, the optimal portfolio is highly diversified, but still very well approximated by the weights obtained by maximizing expression (4) above. (Portfolio weights for all parameter values are available from the author.)

We have shown that the mean-variance approximations are very good for monthly data; however, will they continue to serve as useful surrogates for longer holding periods? Table 4 presents the results of the portfolio simulations using semi-annual data⁴ for the same time period (1960-1976) and the same three samples of

⁴The semiannual returns are calculated as the product of the six monthly returns.

TABLE 2

RATIOS OF EXPECTED UTILITY USING MEAN-VARIANCE-OPTIMAL
WEIGHTS (i.e., WEIGHTS MAXIMIZING EXPRESSION (4))
TO EXPECTED UTILITY USING UTILITY-OPTIMAL WEIGHTS
(i.e., WEIGHTS MAXIMIZING EXPRESSION (5))

Logarithmic $U=\ln(Y+b_1)$	Sample 1	Sample 2	Sample 3	Relative Risk Aversion	Alpha
$b_1=0.0$	.9999732 (4)	.9997950 (4)	.9994121 (3)	1.0	.5
$b_1=0.5$	.9999999 (3)	.9999992 (3)	.9999995 (2)	.67	.333
$b_1=1.0$	.9999999 (3)	.9999999 (2)	.9999996 (2)	.5	.25
$b_1=5.0^a$	1.0 (1)	1.0 (1)	1.0 (1)	.167	.083
Power $U=Y^{b_2}$					
$b_2=.05$	.9999999 (3)	.9999999 (4)	.9999995 (3)	.95	.475
$b_2=0.3$	.9999999 (3)	.9999996 (3)	.9999998 (2)	.7	.35
$b_2=0.5$	.9998499 (3)	.9999997 (2)	.9999995 (2)	.5	.25
$b_2=0.9^a$	1.0 (1)	1.0 (1)	1.0 (1)	.1	.05
Exponential $U=1-\exp^{-b_3 Y}$					
$b_3=3.0$	.9999996 (4)	.9999984 (7)	.9999962 (4)	3.0	1.5
$b_3=2.0$	.9999985 (4)	.9999997 (6)	.9999970 (4)	2.0	1.0
$b_3=1.0$	.9999999 (4)	.9999999 (4)	.9999984 (3)	1.0	.5
$b_3=0.5$	.9999999 (3)	.9999999 (2)	.9999999 (2)	.5	.25
$b_3=0.2^a$	1.0 (1)	1.0 (1)	1.0 (1)	.2	.1

^aFor these values all wealth was invested in the security with the highest mean return.

^bThe numbers in parentheses represent the number of securities in the optimal portfolios. (Recall there are ten securities in each sample.)

TABLE 3
PORTFOLIO COMPARISONS
(MONTHLY DATA)

	Logarithmic	Power	Exponential	Mean-Variance-Formulation (Expression (4))
Relevant Parameter Values	$b_1=0.0$	$b_2=.05^a$	$b_3=1.0$	-
Alpha (α) Value	.5	.475 ^a	.5 (since $w=1$)	.5 ^b .475 ^c
Portfolio Weights				
Sample 1 Security #				
1	.36488	.32473	.34076	.33431
2	.60430	.62179	.60788	.60747
3	.04708	.05349	.04448	.04534
5	.00174	-	.00688	.01288
Sample 2				
3	.05148	.04009	.04556	.03889
6	.23517	.22810	.22360	.21638
8	.64611	.65762	.66093	.67708
10	.06724	.07419	.06991	.06765
Sample 3				
8	.42378	.43007	.41500	.40315
9	.49699	.51411	.48538	.47304
10	.07923	.05582	.09962	.12381
Relevant Parameter Value	$b_1=1.0$	$b_2=.5$	$b_3=.5$	-
Alpha (α) Value	.25	.25	.25 (since $w=1$)	.25
Portfolio Weights				
Sample 1 Security #				
1	.03077	.02855	.02416	.02952
2	.90017	.90403	.89551	.89897
3	.06906	.06742	.08034	.07151
Sample 2				
6	.13627	.14949	.13583	.13389
8	.86373	.85051	.86417	.86611
Sample 3				
8	.25699	.25547	.26953	.27083
9	.74301	.74453	.72917	.73047

aTo obtain an alpha of 0.5, b_2 would have to equal zero which is outside the parameter range.
bFor use in comparisons with logarithmic and exponential portfolios.
cFor use in comparisons with power utility functions.

TABLE 4
PORTFOLIO COMPARISONS
(SEMIANNUAL DATA)

Relevant Parameter Values	Logarithmic	Power	Exponential	Mean-Variance (Expression (4))
Alpha (α) Values	$b_1=0.0$.5	$b_2=0.05$.475	$b_3=1.0$.5	- .5 ^a .475 ^b
Ratio of Expected Utility Using the Mean-Variance-Optimal Weights to Expected Utility Using Utility-Optimal Weights				
Sample 1	.9993127	.9999962	.9999990	
Sample 2	.9947554	.9999874	.9999588	
Sample 3	.9870628	.9999293	.9998191	
Portfolio Weights				
Sample 1 Security #				
1	.24903	.22179	.28100	.29844
2	.63717	.65324	.62001	.61635
5	.11380	.12497	.09899	.08521
Sample 2				
3	.05446	-	.04179	.03586
6	.05696	.05041	-	-
8	.67960	.72299	.71306	.71021
9	.03655	.03140	.07264	.11061
10	.17243	.19520	.17251	.14332
Sample 3				
8	.38510	.37882	.40179	.39895
9	.45831	.49301	.36076	.28646
10	.15659	.12817	.23745	.31459

^aFor use in comparisons with exponential and logarithmic portfolios.

^bFor use in comparisons with power portfolios.

ten securities. Although, as Samuelson's theorem [15] would predict, the approximations are not as good with semiannual data as they are for monthly data, they are, nonetheless, very good indeed. The portfolio weights themselves are again similar for the three utility functions (since parameters were chosen so that the RRA's would be similar), and the portfolio weights obtained by maximizing expression (4) are very similar to the weights obtained by maximizing expected utility for the general functions.

II. Conclusions

We have shown that the mean-variance formulation given in expression (4) provides a very good local approximation to expected utility for more general utility functions using both monthly and semiannual return data.⁵ Thus, the investor can confidently rely on mean-variance optimization, with attitudes toward local changes in portfolio values reflected by the local relative-risk-aversion coefficient. Further evidence of the importance of the Pratt-Arrow risk-aversion coefficient can be found in recent work by J. G. Kallberg and W. T. Ziemba [5, 6]. Their work is based, in part, on a theorem in which it is proved that, under the assumption of jointly-normally-distributed security returns, maximizing expected utility with different global utility functions will generate identical optimal portfolios if certain computed statistics (resembling Pratt-Arrow risk-aversion coefficients) are identical. Using a normal return distribution for ten securities, Kallberg and Ziemba further demonstrate that the optimal portfolios for seven different utility functions (including the quadratic utility function, the expectation of which is a function of means and variances alone) are very similar when global averages of Pratt-Arrow risk-aversion measures are similar. Whereas the analysis in this paper has shown that an expected-utility-maximizing investor could do quite well for himself with a mean-variance formulation based on the local relative-risk-aversion coefficient, Kallberg and Ziemba verify the importance of the Pratt-Arrow risk-aversion measure in global comparisons of expected-utility-maximizing portfolios.

Certainly the results of this study go a long way toward rejuvenating the

⁵We have considered portfolios consisting solely of common stocks, with no possibility for short sale. This is, of course, a very incomplete representation of an individual's actual investment decisions. Whether the approximations are as good for more realistic portfolio decisions and asset distributions remains to be empirically verified.

It is also true that we are implicitly making any assumptions on utility functions and return distributions necessary to reduce the general portfolio problem to a series of single-period myopic decisions involving monthly or semi-annual holding periods.

mean-variance approach and the investment models based upon that approach. The implications, however, are much more general. The approximation derived above does not depend in any restrictive way upon the utility function being approximated. The only relevant parameter is the local relative-risk-aversion coefficient (i.e., the relative-risk-aversion coefficient evaluated at $r=0$). Perhaps the focus of our interest in positive analysis has been misplaced; the particular functional form of an investor's preferences is shown here to be essentially irrelevant in developing a predictive or descriptive model of his (local) decision behavior. What is shown to be important is his particular Pratt-Arrow risk class.

REFERENCES

- [1] Arrow, K. J. "The Role of Securities in the Optimal Allocation of Risk-Bearing." *Review of Economic Studies*, Vol. 31 (April 1964), pp. 91-96.
- [2] _____. *Aspects of the Theory of Risk-Bearing*. Lectures, Helsinki (1964).
- [3] Breiman, Leo. "Investment Policies for Expanding Business Optimal in the Long-Run Sense." *Naval Research Logistics Quarterly*, Vol. 7 (1960), pp. 647-651.
- [4] Hakansson, Nils H. "Capital Growth and the Mean-Variance Approach to Portfolio Selection." *Journal of Financial and Quantitative Analysis*, Vol. 6 (1971), pp. 517-557.
- [5] Kalberg, J. G., and W. T. Ziemba. "Comparisons of Alternative Utility Functions in Portfolio Selection Problems." Faculty of Commerce Working Paper No. 609, University of British Columbia, Vancouver (October 1978).
- [6] _____. "On the Robustness of the Arrow-Pratt Risk Aversion Measure." *Economic Letters*, Vol. 2 (1979), pp. 21-26.
- [7] Levy, H., and H. M. Markowitz. "Approximating Expected Utility by a Function of Mean and Variance." *American Economic Review*, Vol. 69 (June 1979), pp. 308-317.
- [8] Mossin, Jan. "Optimal Multiperiod Portfolio Policies." *Journal of Business*, Vol. 41, No. 2 (April 1968), pp. 215-219.
- [9] Ohlson, James A. "The Asymtotic Validity of Quadratic Utility as the Trading Interval Approaches Zero." In *Stochastic Optimization Models in Finance*, W. T. Ziemba, and R. G. Vickson, eds. New York: Academic Press (1975).
- [10] Ohlson, J. A., and W. T. Ziemba. "Portfolio Selection in a Lognormal Market When the Investor Has a Power Utility Function." *Journal of Financial and Quantitative Analysis* (March 1976), pp. 57-71.
- [11] Pratt, John. "Risk Aversion in the Small and in the Large." *Econometrica*, Vol. 32 (January-April 1964), pp. 122-136.
- [12] Pulley, Lawrence B. "Mean-Variance Approximations to Expected Logarithmic Utility." Working paper.
- [13] _____. "Mean-Variance Approximations to the Growth-Optimal Decision Rule: An Empirical Investigation." Unpublished Ph.D. Thesis, University of Virginia (May 1980).
- [14] Rubinstein, Mark. "The Strong Case for the Generalized Logarithmic Utility Function as the Premier Model of Financial Markets." *Journal of Finance*, Vol. 31 (May 1976), pp. 555-571.
- [15] Samuelson, Paul A. "The Fundamental Approximation Theorem of Portfolio Analysis in Terms of Means, Variances, and Higher Moments." *Review of Economic Studies*, Vol. 36 (October 1970), pp. 537-542.

- [16] Thorp, Edward O. "Portfolio Choice and the Kelly Criteria." In *Stochastic Optimization Models in Finance*, Ziemba and Vickson, eds. New York: Academic Press (1975).
- [17] Tsiang, S. C. "The Rationale of the Mean-Standard Deviation Analysis, Skewness, Preference, and the Demand for Money." *American Economic Review* (June 1972), pp. 354-371. See also comments by Borch and Levy, and reply by Tsiang *AER* (June 1974).
- [18] Young, William E., and Robert N. Trent. "Geometric Mean Approximation of Individual Securities and Portfolio Performance." *Journal of Financial and Quantitative Analysis* (June 1969), pp. 179-199.